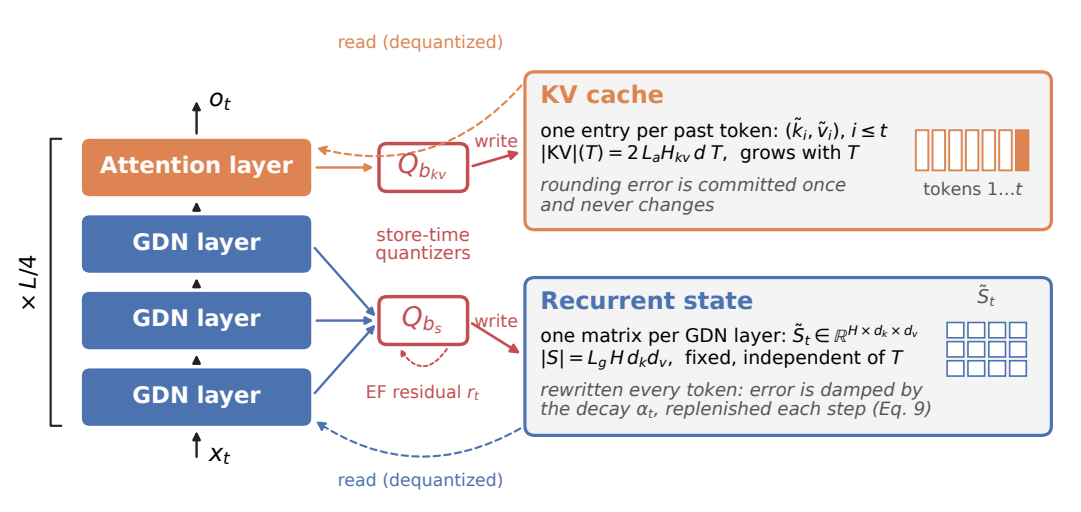
read (dequantized)
o_t
Attention layer
GDN layer
GDN layer
GDN layer
$\times L/4$
x_t
$Q_{b_{kv}}$
write
store-time quantizers
Q_{b_s}
write
EF residual r_t
read (dequantized)
KV cache
one entry per past token: $(\tilde{k}_i, \tilde{v}_i)$, $i \le t$
$|\mathrm{KV}|(T) = 2\,L_a H_{kv}\,d\,T$, grows with T
rounding error is committed once and never changes
tokens 1…t
Recurrent state
$\tilde{S}_t$
one matrix per GDN layer: $\tilde{S}_t \in \mathbb{R}^{H \times d_k \times d_v}$
$|S| = L_g\,H\,d_k d_v$, fixed, independent of T
rewritten every token: error is damped by the decay α_t, replenished each step (Eq. 9)